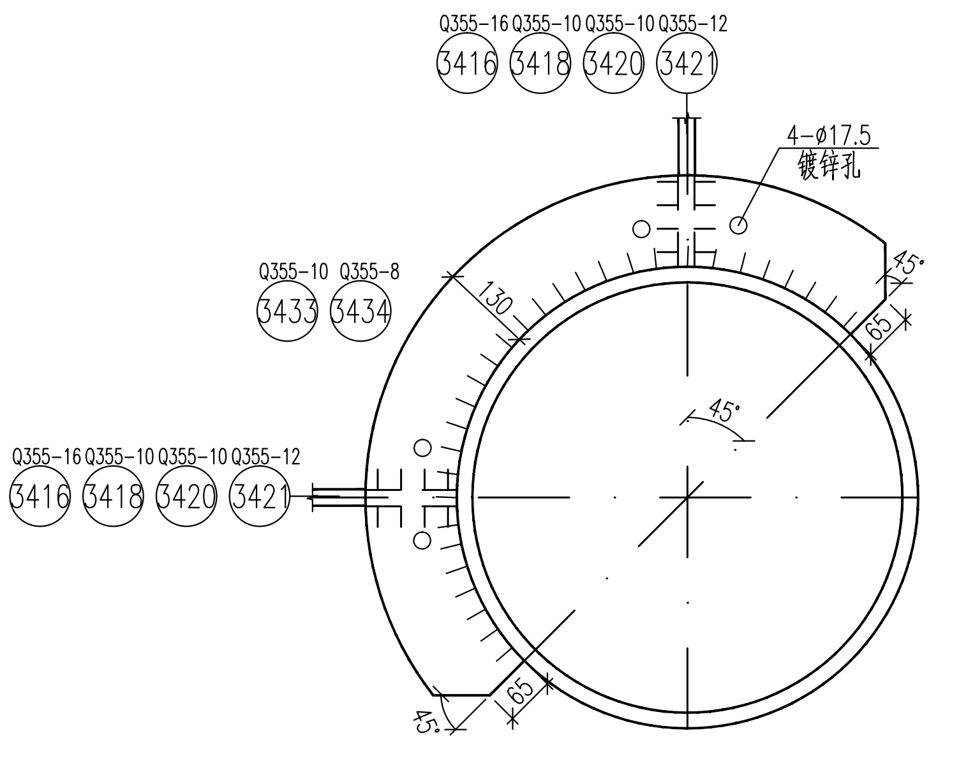
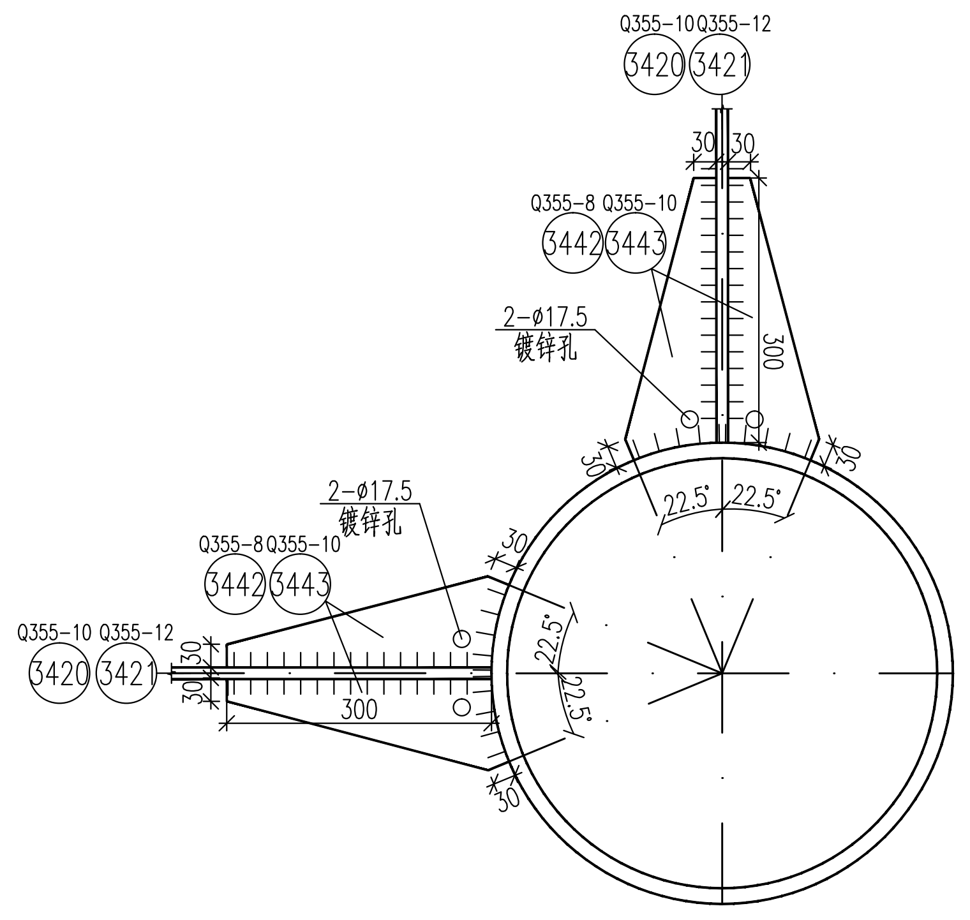
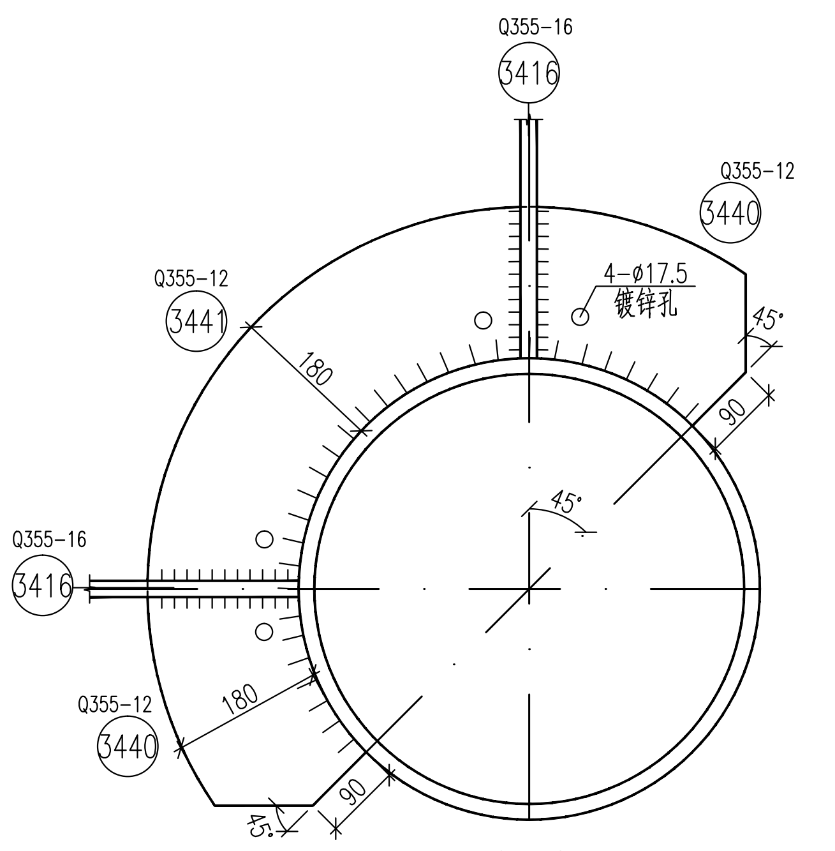


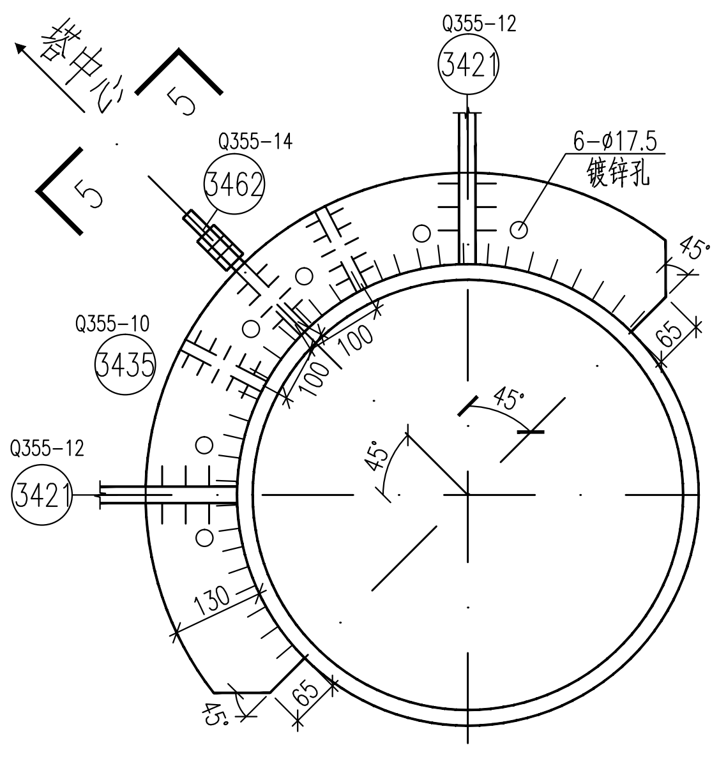
2－2
1:20



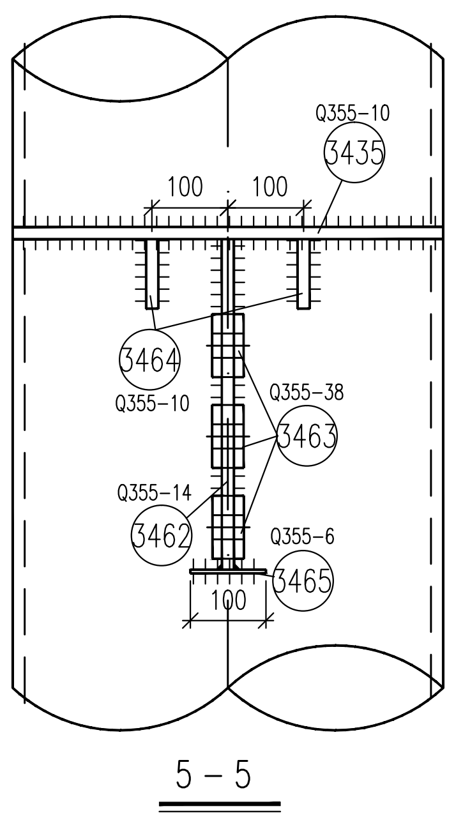
3－3
1:20



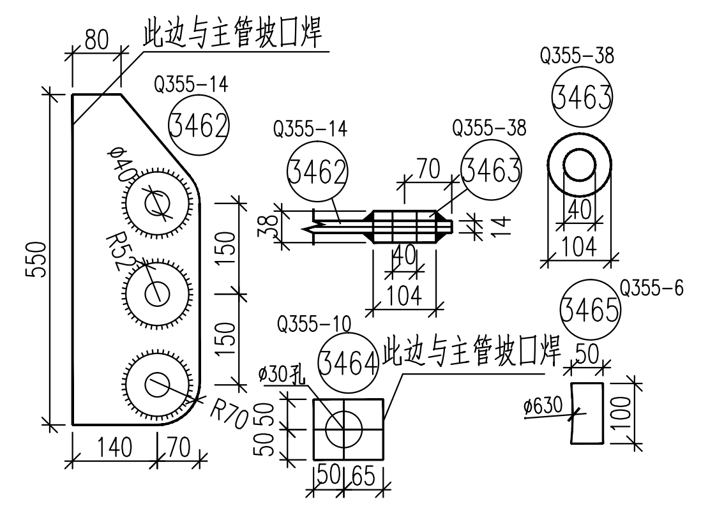
9－9
1:20

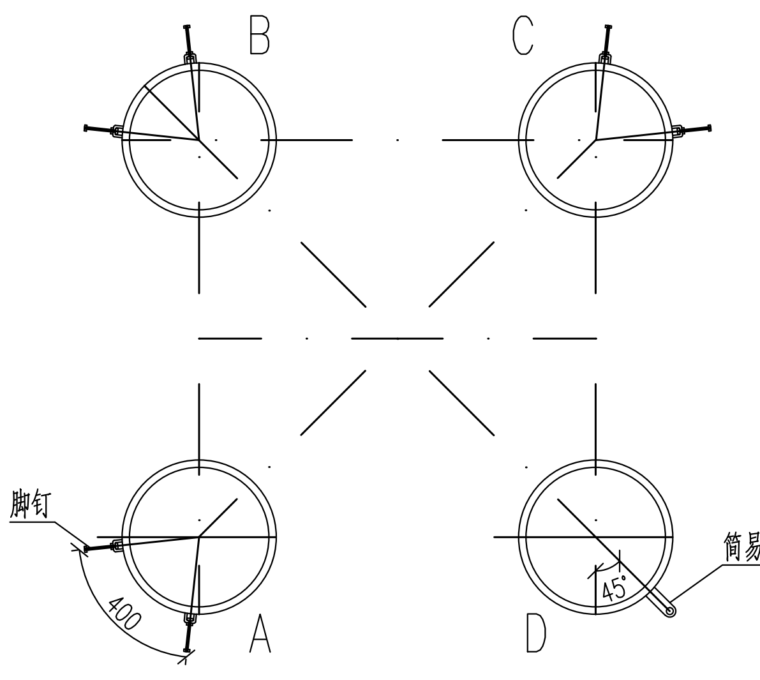


4－4
1:20

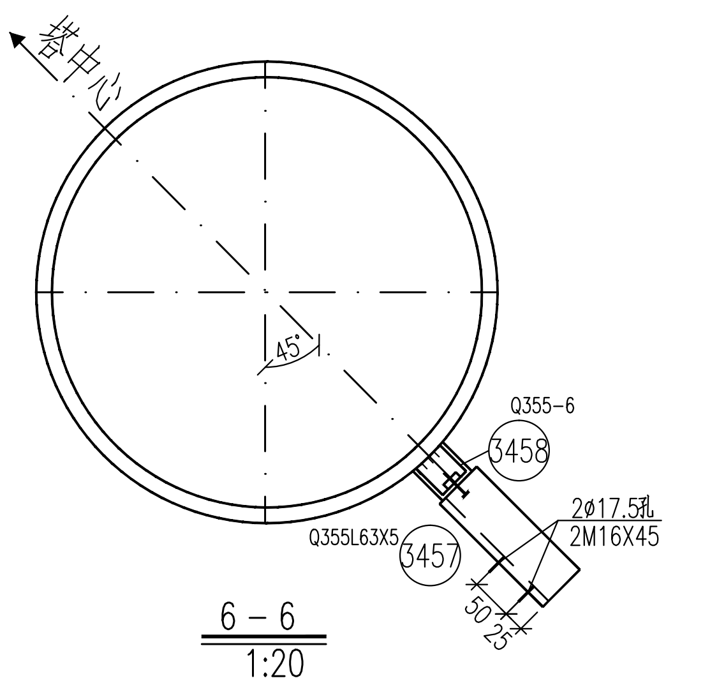


5－5
1:20

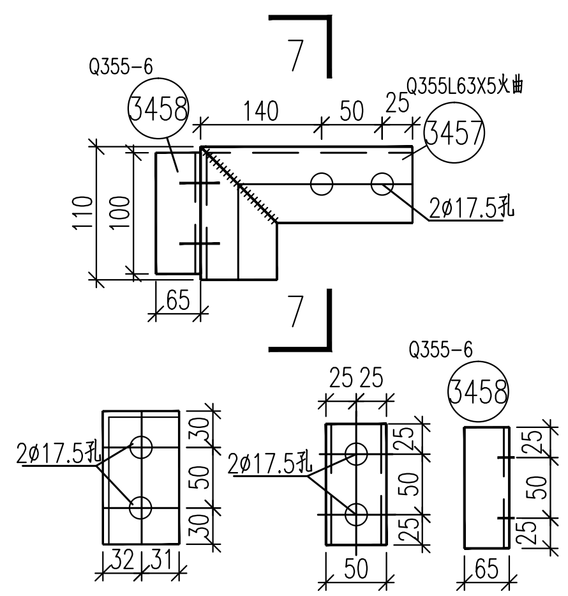




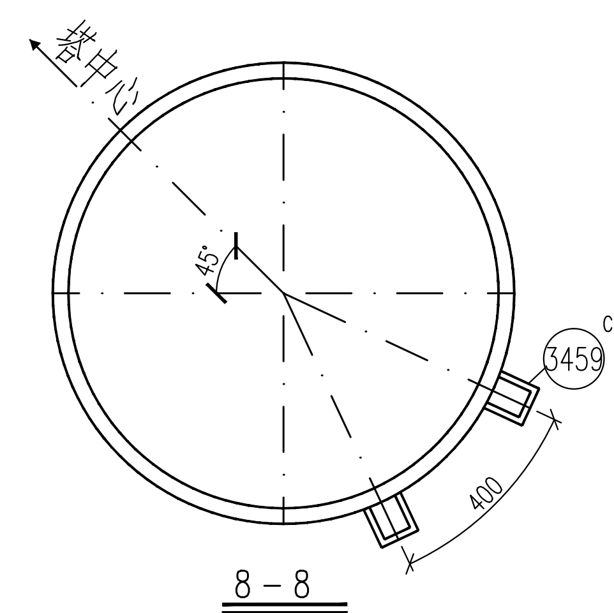
爬梯脚钉示意图



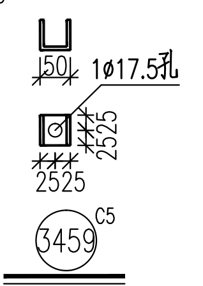
6－6
1:20



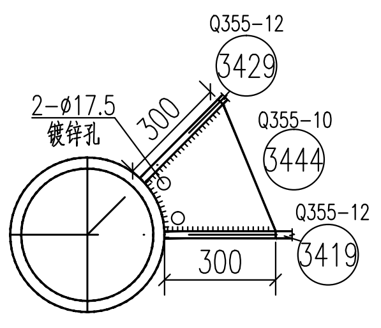
7－7
1:5



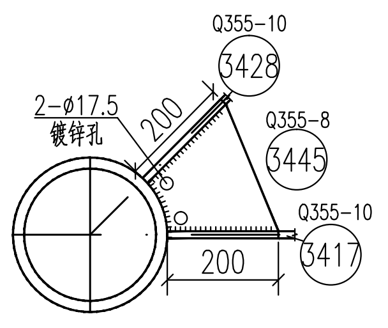
8－8
1:20



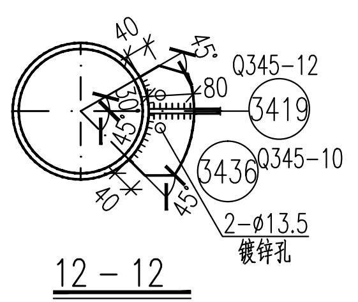
1:20



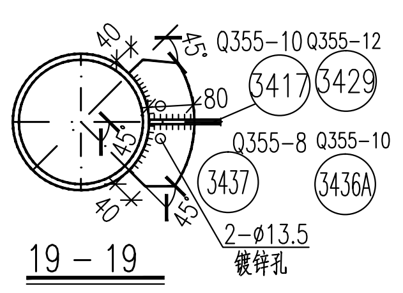
13－13
1:20



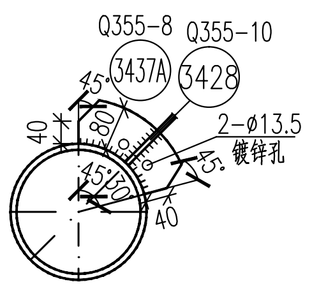
14－14
1:20



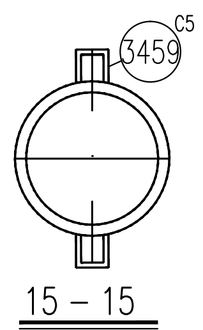
12－12
1:10



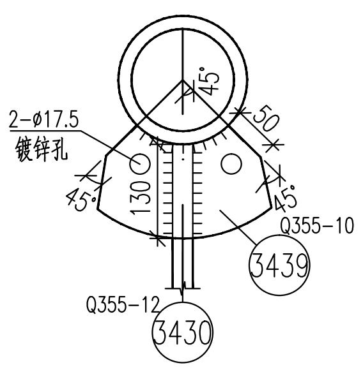
19－19
1:10



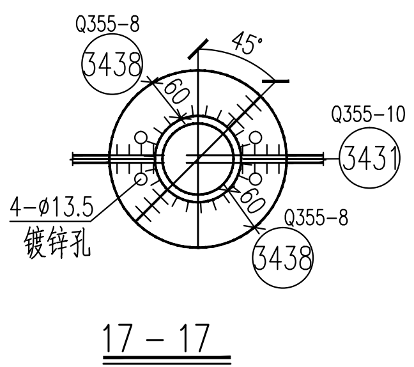
20－20
1:10



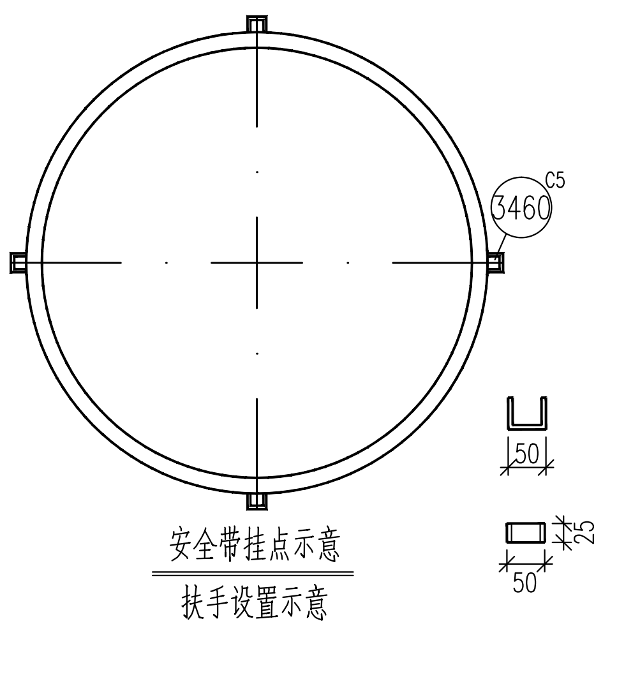
15－15
1:20



16－16
1:20

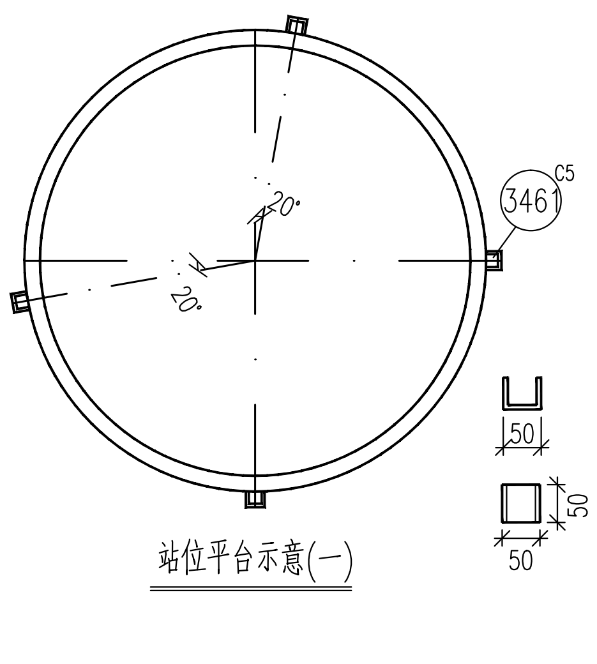


17－17
1:20

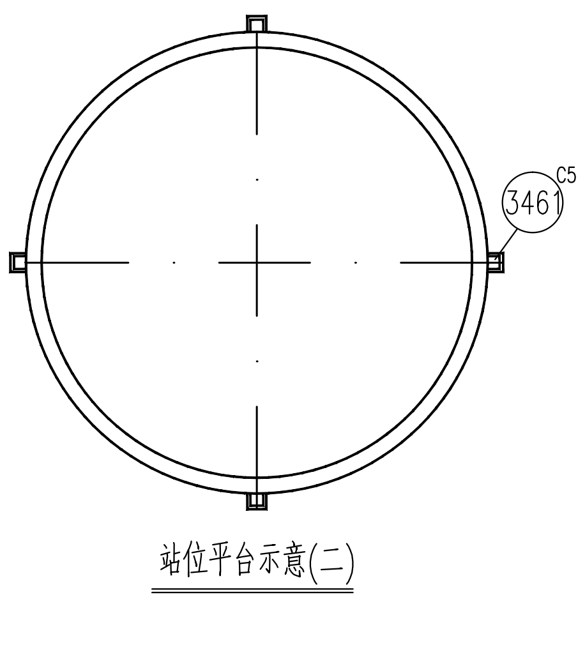


安全带挂点示意

扶手设置示意



站位平台示意(一)



站位平台示意(二)

## 构件明细表

| 编号 | 规格 | 长度(mm) | 数量 | 重量(kg) |  | 备注 |
|---|---|---|---|---|---|---|
|  |  |  |  | 一件 | 小计 |  |
| 3401 | Q420ø630X14 | 1938 | 1 | 412.18 | 412.2 | 一端切弧 带脚钉 |
| 3402 | Q420ø630X14 | 11684 | 1 | 2484.96 | 2485.0 | 装爬梯 |
| 3403 | Q420ø630X14 | 11684 | 0 | 2484.96 | 0.0 | 带脚钉 |
| 3404 | Q420ø630X14 | 11284 | 1 | 2399.89 | 2399.9 | 装爬梯 |
| 3405 | Q420ø630X14 | 11284 | 0 | 2399.89 | 0.0 | 带脚钉 |
| 3406 | Q355ø325X6 | 8546 | 1 | 403.39 | 403.4 | 带脚钉 |
| 3407 | Q355ø325X6 | 8546 | 1 | 403.39 | 403.4 | 带脚钉 |
| 3408 | Q355ø325X6 | 9277 | 1 | 437.89 | 437.9 | 带脚钉 |
| 3409 | Q355ø325X6 | 9277 | 1 | 437.89 | 437.9 | 带脚钉 |
| 3410 | Q355ø325X6 | 7936 | 1 | 374.60 | 374.6 | 带脚钉 |
| 3411 | Q355ø325X6 | 7936 | 1 | 374.60 | 374.6 | 带脚钉 |
| 3412 | Q355ø140X4 | 2828 | 2 | 37.94 | 75.9 |  |
| 3413 | Q355ø203X5 | 7265 | 2 | 177.37 | 354.7 | 带脚钉 |
| 3414 | Q355ø194X5 | 6286 | 2 | 146.50 | 293.0 |  |
| 3415 | Q355ø245X5 | 9026 | 2 | 267.11 | 534.2 | 带脚钉 |
| 3416 | Q355-16X629 | 1815 | 2 | 143.39 | 286.8 |  |
| 3417 | Q355-10X371 | 880 | 2 | 25.63 | 51.3 |  |
| 3418 | Q355-10X193 | 380 | 2 | 5.76 | 11.5 |  |
| 3419 | Q355-12X428 | 814 | 2 | 32.82 | 65.6 |  |
| 3420 | Q355-10X453 | 1056 | 2 | 37.55 | 75.1 |  |
| 3421 | Q355-12X485 | 1249 | 2 | 57.06 | 114.1 |  |
| 3422 | Q355ø140X4 | 4492 | 1 | 60.26 | 60.3 |  |
| 3423 | Q355ø180X5 | 11037 | 1 | 238.17 | 238.2 | 带脚钉 |
| 3424 | Q355ø180X5 | 3221 | 1 | 69.51 | 69.5 | 带脚钉 |
| 3425 | Q355ø180X5 | 6996 | 1 | 150.97 | 151.0 | 带脚钉 |
| 3426 | Q355ø245X5 | 9407 | 1 | 278.39 | 278.4 | 带脚钉 |
| 3427 | Q355ø245X5 | 9407 | 1 | 278.39 | 278.4 | 带脚钉 |
| 3428 | Q355-10X295 | 627 | 2 | 14.52 | 29.0 |  |
| 3429 | Q355-12X446 | 1012 | 2 | 42.52 | 85.0 |  |
| 3430 | Q355-12X497 | 990 | 1 | 46.35 | 46.4 |  |
| 3431 | Q355-10X570 | 484 | 1 | 21.66 | 21.7 |  |
| 3432 | Q355-10X570 | 231 | 2 | 10.34 | 20.7 |  |
| 3433 | Q355-10X130 | 1313 | 2 | 13.40 | 26.8 |  |
| 3434 | Q355-8X130 | 1313 | 4 | 10.72 | 42.9 |  |
| 3435 | Q355-10X130 | 1313 | 1 | 13.40 | 13.4 |  |
| 3436 | Q355-10X65 | 255 | 4 | 1.30 | 5.2 |  |
| 3436A | Q355-10X65 | 306 | 4 | 1.56 | 6.2 |  |
| 3437 | Q355-8X65 | 255 | 4 | 1.04 | 4.2 |  |
| 3437A | Q355-8X65 | 306 | 4 | 1.25 | 5.0 |  |
| 3438 | Q355-8X60 | 415 | 4 | 1.56 | 6.2 |  |
| 3439 | Q355-10X130 | 305 | 2 | 3.11 | 6.2 |  |
| 3440 | Q355-12X180 | 306 | 4 | 5.19 | 20.8 |  |
| 3441 | Q355-12X180 | 613 | 2 | 10.39 | 20.8 |  |
| 3442 | Q355-8X300 | 201 | 4 | 3.79 | 15.2 |  |
| 3443 | Q355-10X300 | 201 | 4 | 4.73 | 18.9 |  |
| 3444 | Q355-10X300 | 330 | 2 | 7.77 | 15.5 |  |
| 3445 | Q355-8X200 | 220 | 2 | 2.76 | 5.5 |  |
| 3446 | -16X845 | 845 | 1 | 70.43 | 70.4 | 遮水板 |
| 3447 | Q355-8X80 | 110 | 1 | 0.55 | 0.6 |  |
| 3448 | Q420-32X1316 | 1316 | 1 | 263.37 | 263.4 |  |
| 3449 | Q355-16X343 | 390 | 1 | 16.80 | 16.8 |  |
| 3450 | Q355-16X343 | 390 | 2 | 16.80 | 33.6 |  |
| 3451 | Q355-16X343 | 390 | 2 | 16.80 | 33.6 |  |
| 3452 | Q355-16X343 | 390 | 2 | 16.80 | 33.6 |  |
| 3453 | Q355-16X343 | 390 | 2 | 16.80 | 33.6 |  |
| 3454 | Q355-16X343 | 390 | 2 | 16.80 | 33.6 |  |
| 3455 | Q355-16X343 | 390 | 2 | 16.80 | 33.6 |  |
| 3456 | Q355-16X343 | 390 | 1 | 16.80 | 16.8 |  |
| 3457 | Q355L63X5 | 325 | 17 | 1.57 | 26.7 | 火曲（用于主材装爬梯） |
| 3458 | Q355-6X100 | 180 | 17 | 0.85 | 14.5 | 火曲（用于主材装爬梯） |
| 3459 | C5 | 50 | 205 | 0.27 | 55.4 | 槽钢（用于主材装爬梯） |
| 3459 | C5 | 50 | 266 | 0.27 | 71.8 | 槽钢（用于主材带脚钉） |
| 3460 | C5 | 25 | 12 | 0.14 | 1.7 | 槽钢 |
| 3461 | C5 | 50 | 8 | 0.27 | 2.2 | 槽钢 |
| 3462 | Q355-14X210 | 550 | 1 | 12.69 | 12.7 |  |
| 3463 | Q355-38X104 | 104 | 3 | 3.23 | 9.7 |  |
| 3464 | Q355-10X100 | 115 | 2 | 0.90 | 1.8 |  |
| 3465 | Q355-6X50 | 100 | 1 | 0.24 | 0.2 |  |
| 3467 | Q355-12X80 | 760 | 2 | 5.73 | 11.5 |  |
|  | C1404H5 |  | 6 | 4.80 | 28.80 |  |
|  | C1805H5 |  | 4 | 7.12 | 28.48 |  |
|  | C2405H5 |  | 8 | 17.27 | 138.16 |  |
|  | C1905H5 |  | 4 | 8.80 | 35.20 |  |
|  | C2005H5 |  | 4 | 9.74 | 38.96 |  |
|  | C3206H5 |  | 4 | 31.43 | 125.72 |  |
|  | X1805H7 |  | 2 | 21.60 | 43.20 |  |
|  | FD3232B7H |  | 8 | 16.80 | 134.4 |  |
|  | FD6363B7D |  | 5 | 103.00 | 515.0 |  |
|  | T3930 | – | 2 | 20.70 | 20.70 | 主材安装爬梯 |
|  | T6980 | – | 2 | 36.20 | 108.60 | 主材安装爬梯 |
|  | T2330 | – | 1 | 12.40 | 12.4 | 主材安装爬梯 |
|  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |
| 合计 |  | 12851.6 kg(用于主材带脚钉) |  |  |  |  |
| 合计 |  | 13018.1 kg(用于主材装爬梯) |  |  |  |  |

## 螺栓、脚钉、垫圈明细表

| 名称 | 级别 | 规格 | 符号 | 数量 | 重量(kg) | 备注 |
|---|---|---|---|---|---|---|
| 螺栓 | 6.8级 | M16X45 | ◎ | 68 | 11.5 | 用于主材装爬梯 |
|  |  | M16X65 | ⊗ | 48 | 9.6 |  |
|  |  | M20X65 | ⌀ | 48 | 16.9 |  |
|  | 8.8级 | M24X75 | ⊘ | 66 | 39.1 |  |
|  |  | M20X100 | ○ | 88 | 43.6 | 法兰螺栓 |
|  |  | M39X175 | ○ | 52 | 184.6 | 法兰螺栓 |
|  |  | M39X190 | ○ | 26 | 96.0 | 法兰螺栓 |
| 脚钉 | 6.8级 | M16X185 | ⊖ ⊣ | 205 | 85.0 | 用于主材装爬梯 |
|  |  | M16X185 |  | 266 | 110.3 | 用于主材带脚钉 |
|  |  |  |  |  |  |  |
| 垫圈 | Q235 |  | 规格X个数 |  |  |  |
|  |  |  |  |  |  |  |
| 合计 |  |  | 500.1 kg (用于主材带脚钉) |  |  |  |
| 合计 |  |  | 486.3 kg (用于主材装爬梯) |  |  |  |

说明：1.材料表为一个塔腿的重量。

2.本段塔腿为铁塔D腿，安装爬梯，当为A、B、C腿时需焊接脚钉连接槽钢。

**山东电力工程咨询院有限公司**
浙江特高压交流环网工程线路 工程
施工图 设计阶段

| 批准 |  | 校核 |  |
|---|---|---|---|
| 审核 | 高锡鹏 |  |  |
|  |  | 设计/勘测 |  |
| 比例 |  | 日期 | 2026年 08月 |

SZK31102A双回路直线塔
96.0－0.0m塔腿结构图(3/3) ㉞

图号 37-SA36901S-T0121-074 版本